

ENGINEERING BULLETIN

Product Options

Fanstech fans MoDscan activities

No: EB/23-24/003,
10/08/2023

Dear Colleague,

We here by wish to inform you about the official release of Modbus configuration procedure for our new developed data center chillers fans, we are using FANSTECH makes fans for air cooled condenser, That fans has inbuilt VFD for smooth operation.

We will cover below topics in this document for Fanstech Fans.

- Fans Addressing.
- Fans Fault Finding.
- Fans Configuration.
- Fans manual Testing.

MOD SCAN ACTIVITIES

- Steps for MoDscan
 1. Download MoDscan
 2. Start-up of the Mod Scan Software
 3. Process to Selection Comport
 4. Action activities with Software
 5. Testing the Module with Mod Scan Software
 6. Testing report after action

How to Download MoDscan Software: -

Open any search engine and go to this link--modscan32.zip

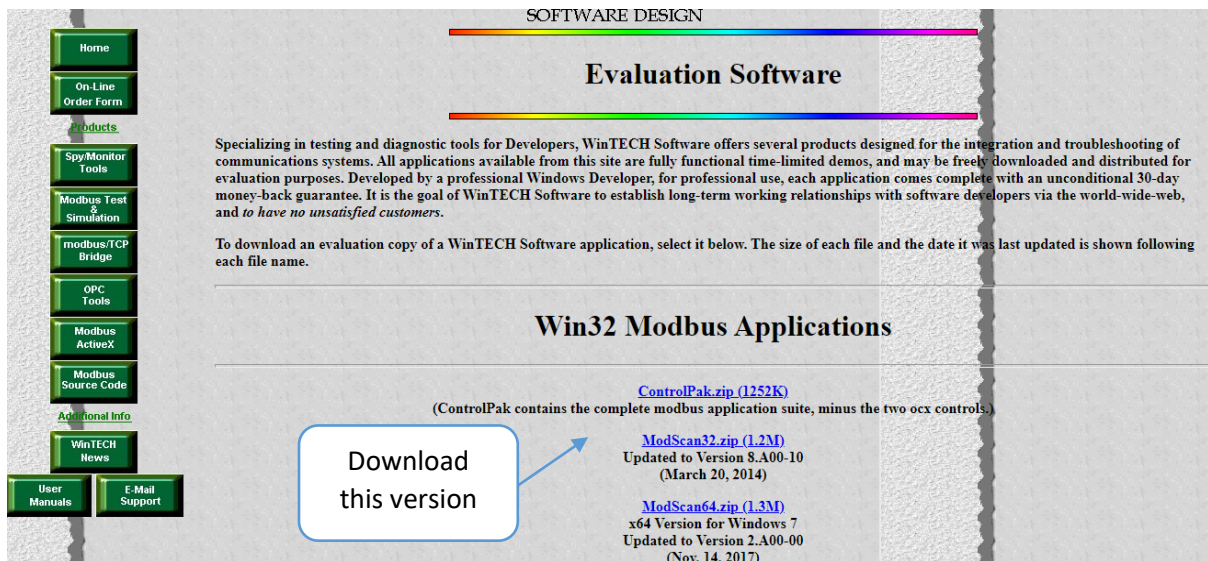


Fig 1. Modscan Software

Note:-Download MoDscan32.zip and Extract this file.

- Start-up of the ModScan Software

After extraction the ModScan Zip files have a setup file (Setup file doesn't need to installation in PC). Open setup and ignore all the login section by the cancellation and its ready for testing.

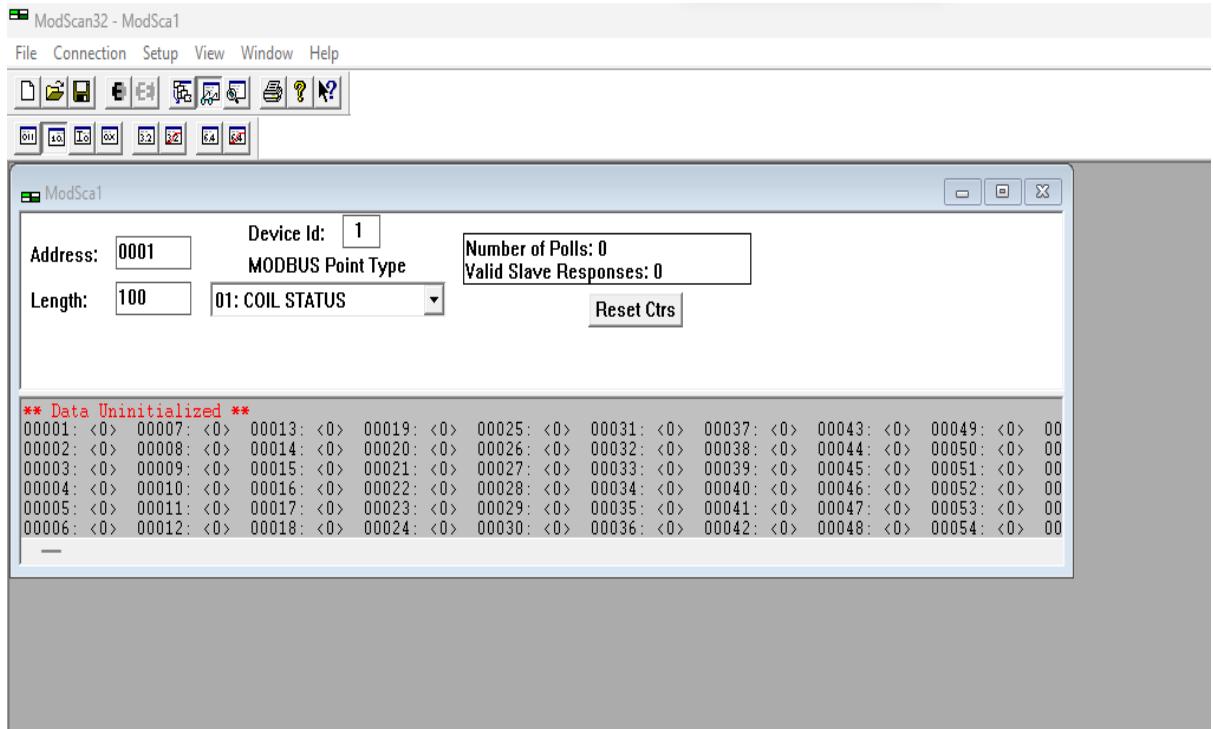


Fig 2. Start-up of Modscan Software

- Process of ports selection

For Selection of port selection setting, go to the window and search device manager.

Ports (Com & LPT)

See the com port with number.

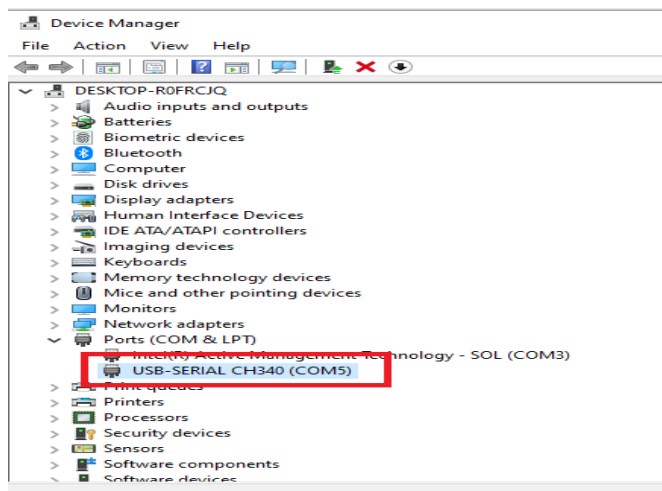


Fig 3. Com port selection

Select the connect window in MoDscan software and selection of com port and configuration by default.

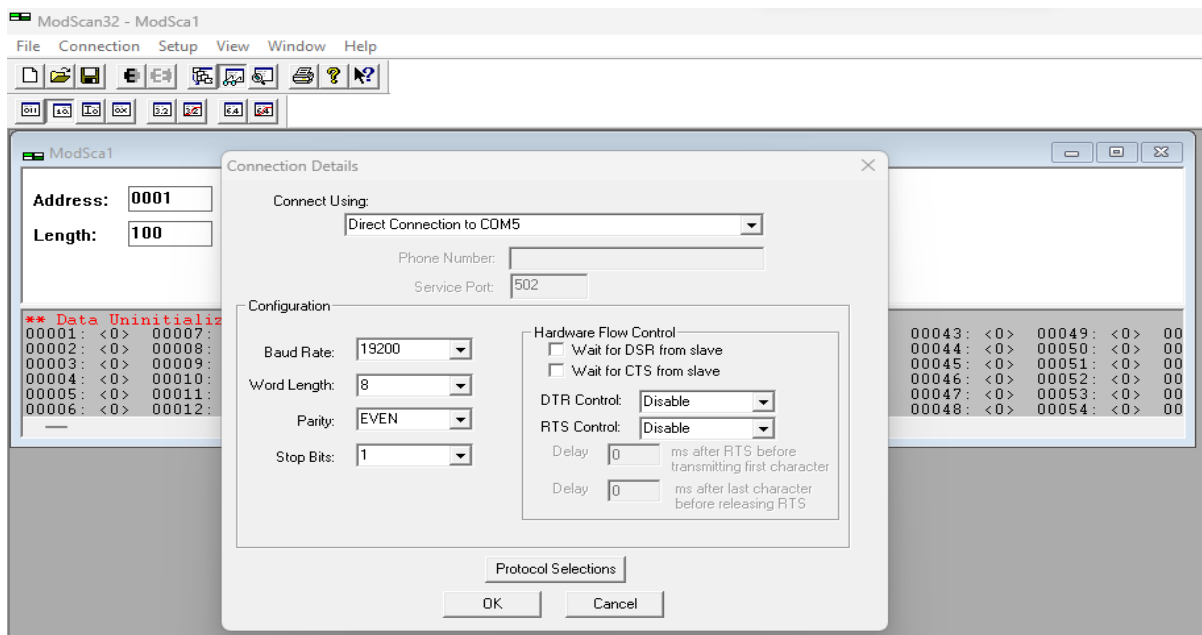


Fig 4. Com port selection

- Action activities with Software
Refer the fan documents for communication addresses.
In fan we need to update the 3-basic parameters.
 1. Fan Communication Address: -53505.
 2. Fan Speed Control Mode Address: -53506.
 3. Fan Speed (RPM) Address: -53250.

Create the 3 window in the MoDscan Software for these above mention address.

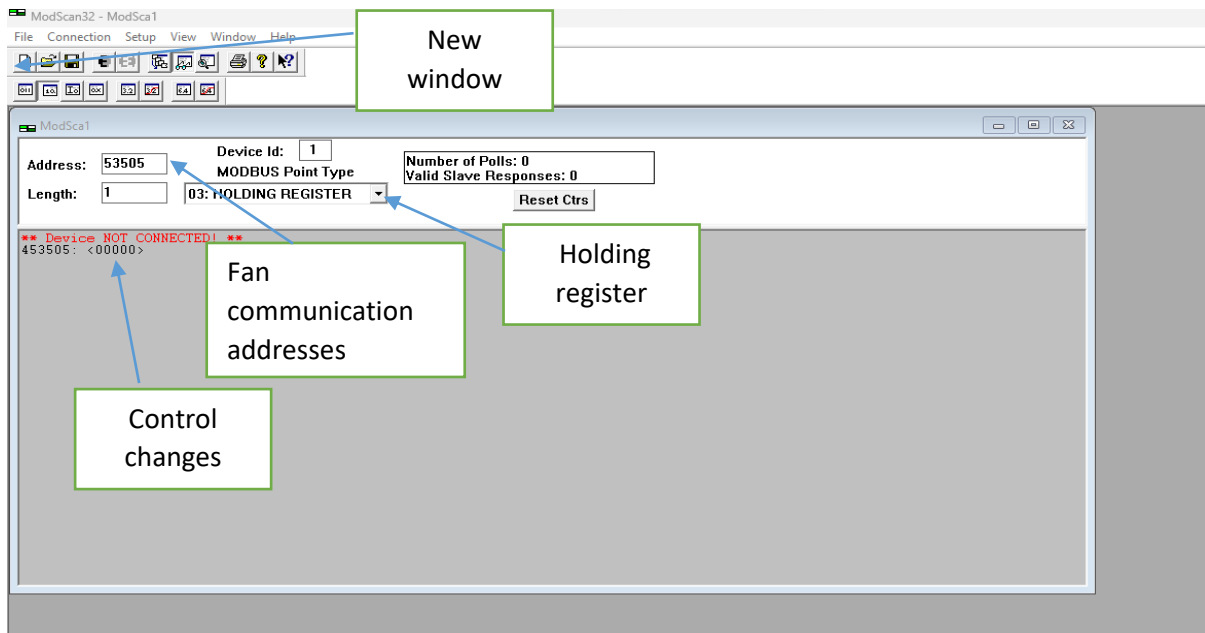


Fig 5. New Address MoDscan Window

Testing the Module with MoDscan Software create 3 windows address only and take action accordingly.

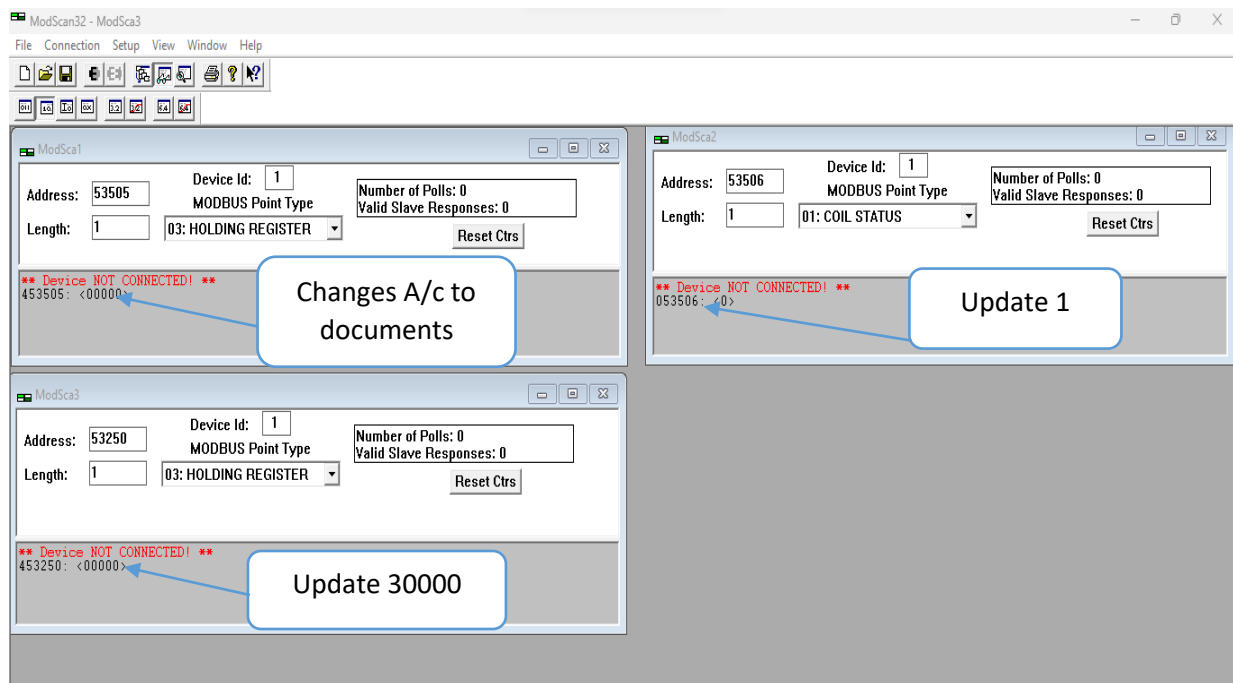


Fig 6. Module Addresses Windows

Updates in all 3 parameters according to the mention fan with location.

In fan we need to update the 3-basic parameters.

1. Fan Communication Address: -53505.

Changes the fan address according to the fan documents.

2. Fan Speed Control Mode Address: -53506.

Set: -00 Analog Speed Regulation Mode

Set: -02 Communication Speed Regulation modes

3. Fan Speed (RPM) Address: -53250.

This is the fan speed parameter according to this rpm fan got on and rpm both.

Changes the parameters according to the fan documents and take a power cycle off-on condition.

After the power off-on cycle take the online with the MoDscan accordingly and check the fan on-off condition by the putting the rpm in 53250 by 30000 if fan got on by this value check physically and after checking on again put 0 in same then fan got off command.

If this both fan got on-off by the command, then MoDscan activities done for the particular fan.

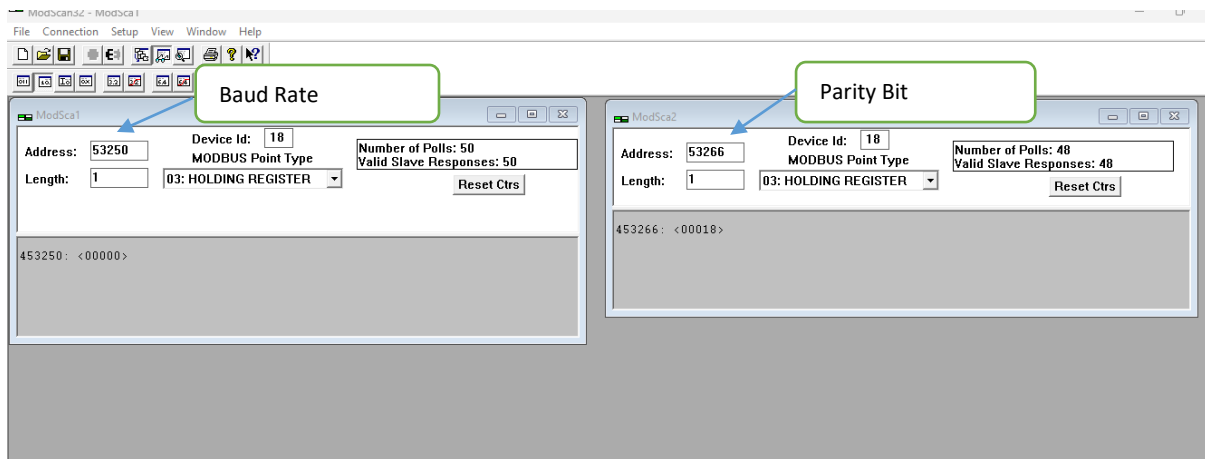
All Communication Addresses with Descriptions

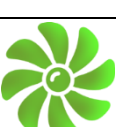
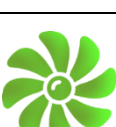
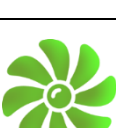
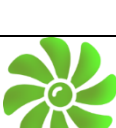
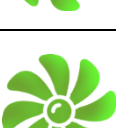
The screenshot displays six Modbus communication windows in the ModScan32 software, each with a green callout box identifying a specific data point:

- ModSca1:** Address 53249, Device Id 18, MODBUS Point Type, Number of Polls: 5, Valid Slave Responses: 5. Callout: **Control Word Reset**.
- ModSca2:** Address 53250, Device Id 18, MODBUS Point Type, Number of Polls: 11, Valid Slave Responses: 11. Callout: **Speed(RPM)**.
- ModSca3:** Address 53265, Device Id 18, MODBUS Point Type, Number of Polls: 6, Valid Slave Responses: 6. Callout: **Current Actual Speed**.
- ModSca4:** Address 53266, Device Id 18, MODBUS Point Type, Number of Polls: 5, Valid Slave Responses: 5. Callout: **Fault Word**.
- ModSca5:** Address 53270, Device Id 18, MODBUS Point Type, Number of Polls: 6, Valid Slave Responses: 6. Callout: **IPM Module Temperature**.
- ModSca6:** Address 53275, Device Id 18, MODBUS Point Type, Number of Polls: 7, Valid Slave Responses: 7. Callout: **Query Set Speed**.

The screenshot displays six Modbus communication windows in the ModScan32 software, each with a green callout box identifying a specific data point:

- ModSca1:** Address 53282, Device Id 18, MODBUS Point Type, Number of Polls: 102, Valid Slave Responses: 62. Callout: **Control Output Power**.
- ModSca2:** Address 53505, Device Id 18, MODBUS Point Type, Number of Polls: 202, Valid Slave Responses: 123. Callout: **Fan Address**.
- ModSca3:** Address 53506, Device Id 18, MODBUS Point Type, Number of Polls: 102, Valid Slave Responses: 62. Callout: **Speed Mode**.
- ModSca4:** Address 53508, Device Id 18, MODBUS Point Type, Number of Polls: 102, Valid Slave Responses: 62. Callout: **Control Value Storage**.
- ModSca5:** Address 53530, Device Id 18, MODBUS Point Type, Number of Polls: 104, Valid Slave Responses: 62. Callout: **Max Speed**.
- ModSca6:** Address 53531, Device Id 18, MODBUS Point Type, Number of Polls: 139, Valid Slave Responses: 65. Callout: **Limited Speed**.



<u>PANEL</u>	
2  2	1  1
6  3	5  4
10  5	9  6
14  7	13  8
18  9	17  10
20  2	19  1
16  3	15  4
12  5	11  6
8  7	7  8
4  9	3  10

Modbus addresses
for FAN